# RAMCO INSTITUTE OF TECHNOLOGY

**Rajapalayam**

**Department of Computer Science and Engineering**
**Academic Year: 2019- 2020 (Even Semester)**

**Degree, Semester & Branch: VI Semester B.E. Computer Science and Engineering.**
**Course Code & Title: CS8601 Mobile Computing**
**Name of the Faculty member: Dr.M.Kaliappan, Associate Professor/CSE**

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**Virtual Lab: Simulation of Mobile Ad-hoc Networks**
**Date and Time: 12.02.2020 & 9.50 am to 10.40am**
**Venue: CC-II**

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**1. Outcomes**

**After completing this experiment, the students will be able to:**

- Describe the basics of Mobile Ad-hoc Networks (MANETs) and different routing protocols
- Setup a network with wireless nodes using ns2

**2. Network simulator.**

Network simulator2 (NS2) designed to simulate and design mobile adhoc network that view in graphically. It shows the network traffic (packet movement) visually and how the packets transferred from source node to destination node. Students can visualize the mobility of the nodes for MANET simulation. We can design both wired and wireless networks with various network protocols such as DSR, AODV, and DSDV etc. Also, we perform analytical studies using network performance metrics such as packet delivery ratio, end-to-end delay and throughput. This gives research insights and motivation to do research in this domain. We can install NS2 Linux environment.

The following link is used to access the virtual lab.

http://vlab.co.in/ // URL for Virtual lab

http://vlabs.iitkgp.ernet.in/ant/ // URL for MANET simulation

This virtual lab is designed and implemented by IIT karagpur, India. This simulation supports various computer networks such as Local Area Network , Satellite Network, Wi-Fi Network, WiMAX Network, Mobile Ad-hoc Network, , Wireless Sensor Network, Bluetooth Network and ZigBee Network.

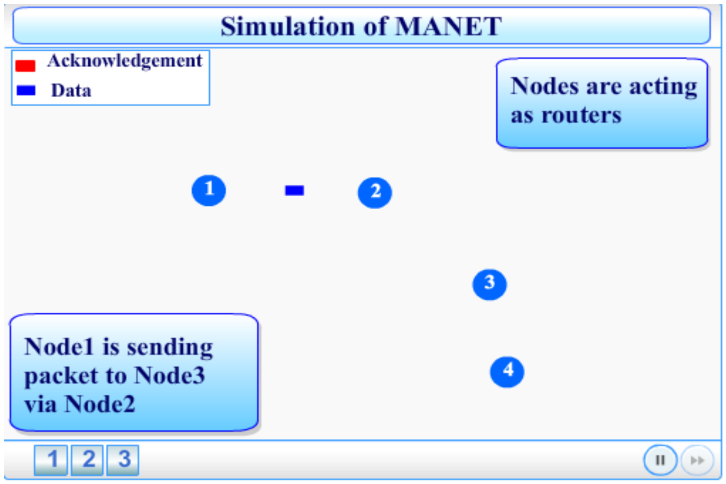
## 3. Manual

### 3.1 Introduction.

The prime goal of this simulation is setup MANET with various routing protocols like AODV, DSR, and DSDV. The simulation is written by TCL script [2]. The network performance is analyzed by trace file. The trace file is generated automatically after the simulation is completed. The trace file contains all activities happened during the simulation and it recorded all the data based on the event. This simulation shows the MANET setup, how a packet is transferred from source to destination, what happen if the network gets disrupted, and how new routing path is constructed after disruption or topology changes or node dead. This simulation is setup with four mobile nodes and performs the packet transfer and acknowledgement, route disruption and new route formation. This simulation contains introduction, theory, simulation, self-evaluation, procedure, exercises and references section to get the insight of routing in MANET.

### 3.2 Instructions to perform a simulation experiment.

- **Visit to the following URL to perform a simulation for Routing in MANET.**
  http://vlabs.iitkgp.ernet.in/ant/7/
- This page contains five tabs such as Introduction, theory, Simulation, Self-evaluation, procedure, exercise and references to perform the MANET simulation.
- Select the simulation tab.
- Click the button 1 to setup a MANET with four nodes. It shows how nodes form a network automatically and performs packet transformation from node1 to node 4 with acknowledgement. Figure 1 shows the simulation view.

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- Figure1 : Simulation view of four nodes

- Then, click button 2. It describe and show a packet transformation when disruption of routing path between node 1 to node2. Here, when node2 moved out of the coverage, new routing path will be constructed between node 1 and node 2 without node2. Figure 2 shows the route disruption scenario.

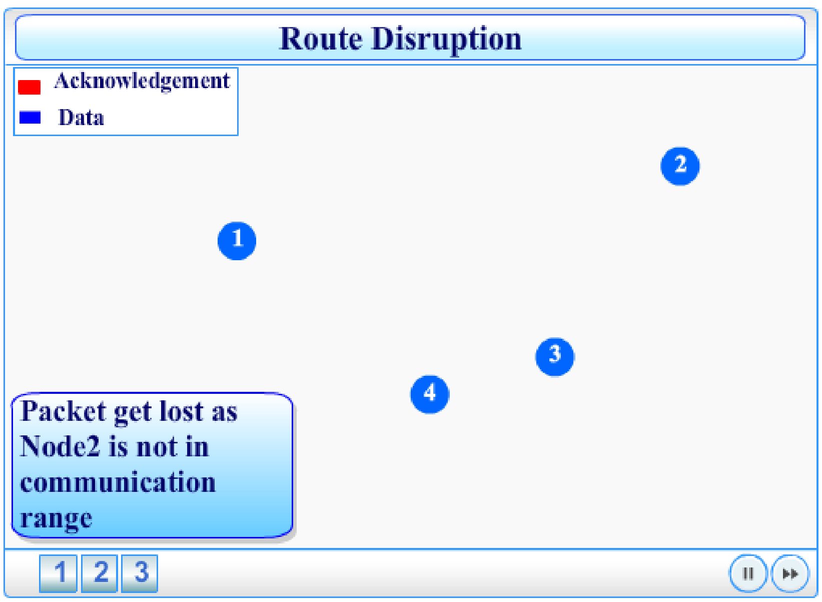



Figure 2: Node mobility.

New routing path (path: Node1 – node 4 – node 3) will be constructed anutomatically when node2 move out of the transmission range. Figure 3 shows the packet transformation through new routing path and figure 4 shows the how acknowledgement packet transfer from destination node(node3) to source node (node1).

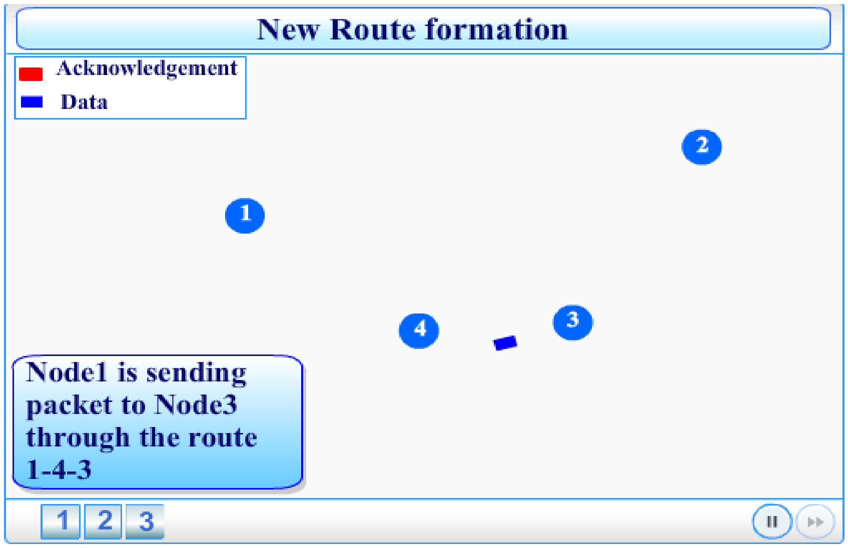



Figure 3: Packet transformation through new route.

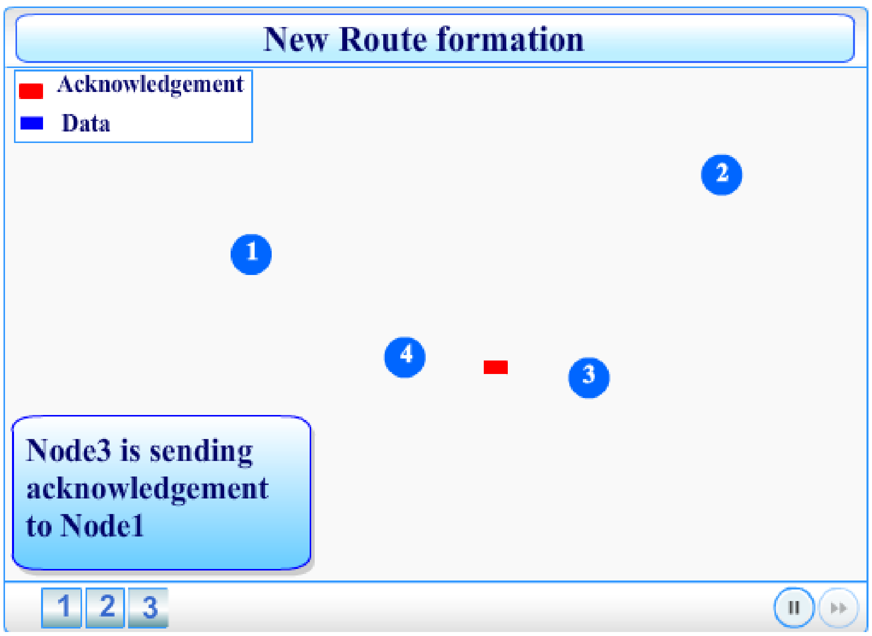



Figure 4: Acknowledgement packet transformation

## 3.3 NS-2 INSTALLATION

**NS2 can be installed in Linux environment [2]. Installation steps are given below.**

- Copy ns-allinone-2.34.tar.gz package to /root folder
- Right click & do extract here
    - It will create ns-allinone-2.34 folder
- Open the terminal and go to ns-allinone-2.34 folder
    - cd /root/ ns-allinone-2.34
- To install type the command
    - ./install
- Open another terminal : set the environment variable open the /etc/profile file
    - gedit /etc/profile
- Enter these variable in end of profile
    - export PATH=$PATH
    - export LD_LIBRARY_PATH=
    - export TCL_ LIBRARY=

Copy the value of PATH, LD_LIBRARY_PATH, and TCL_LIBRARY environment variables and paste it in /etc/profile file.

### 3.3.1 Example:

- set the environment variable
    - export PATH=$PATH:/root/ns-allinone- 2.34/tcl8.4.15/unix:/root/ns-allinone-2.34/tk8.4.15/unix: /root/ns-allinone-2.34/bin
    - export LD_LIBRARY_PATH=/root/ ns-allinone-2.34/lib: /root/ns-allinone-2.34/otcl-1.13
    - export TCL_ LIBRARY=/root/ns-allinone-2.34/tcl8.4.15/library

## 3.4 Frequent Ask Questions and answers.

1. How nodes can be increased in simulation?
2. How can set the node property and routing protocol in same MANET scenario?
3. How can we analyse the performance of the MANET?
4. How can develop, design and implement new protocol for our research?

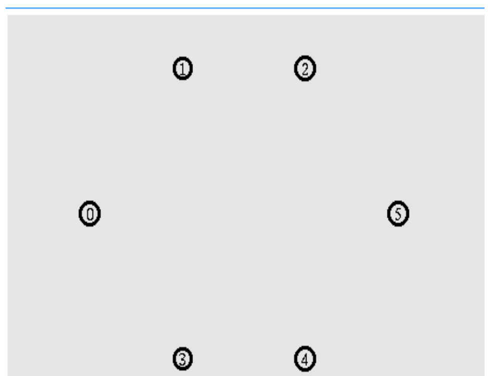
### 3.4.1 Answer

**1. How nodes can be increased in simulation?**

We write a TCL script in NS2 for MANET simulation. We can increase number of nodes using the following scripts.

```
#Create six nodes
for {set i 0} {$i < 6} {incr i} {
        set n($i) [$ns node]
}

#Create links between the nodes
for {set i 0} {$i < 6} {incr i} {
        $ns duplex-link $n($i) $n([expr ($i+1)%7]) 1Mb 10ms
DropTail
}
```




**#Code for creating specific number of nodes**

```
for {set i 0 } {$i < val(nn)}{incr i} {
     set node_($i) [$ns_ node]
}
```

**#Define a node position randomly**

```
for {set i 0}{$i < $val(nn)}{incr i}{
       $ns_  initial_node_pos $node_($i) 30
}
```

**2. How can set the node property and routing protocol in same MANET scenario?**

The following TCL scripts used to node setting and routing protocol for MANET.

**# node configuration**

```
$ns_   node-config     - adhocRouting AODV\
```

```
- llType LL \
- macType Mac/802_11 \
- ifqType $qtype \
- ifqLen 50 \
- antType Antenna/OmniAntenna \
-propType Propagation/TwoRayGround \
- phyType Phy/WirelessPhy \
- channelType Channel/WirelessChannel\
- topoInstance $topo \
- agentTrace ON \
- routerTrace ON \
- macTrace OFF
```

**3. How can we analyse the performance of the MANET?**

The performance metrics such as hop-count, packet delivery ratio and end-to-end delay used to analyses the MANET performance. The simulation result provides a trace file that contains the activities of each node based on the event. The trace file format is as follows.

s –t 163.001503520 –Hs 0 –Hd -2 –Ni 0 –Nx 300.00 –Ny 500.00 –Nz 0.0 –Ne -1.000000 –Nl AGT –Nw --- –Ma 0 –Md 0 –Ms 0 –Mt 0 –Is 0.0–Id 2.0 –It cbr –Il 200 –If 1 –Ii 77 –Iv 32 –Pn cbr –Pi 32 –Pf 0 –Po 0

Field 0: Event type

- s:send r:receive d:drop f:forward

Field 1: General tag

- -t: time

Field 3: Next hop info

- -Hs: id for this node
- -Hd: id for next hop towards the destination

Field 4: Node property type tag

- Ni: node id
- Nx –Ny –Nz: node’s x/y/z coordinate
- Ne: Node energy level
- Nl: trace level, such as AGT, RTR, MAC
- Nw: reason for the event

Field 5: Packet info at MAC level

❖ Ma: duration
❖ Md: dest's Ethernet address
❖ Ms: src's Ethernet address
❖ Mt: Ethernet type

Field 6: Packet information at IP level

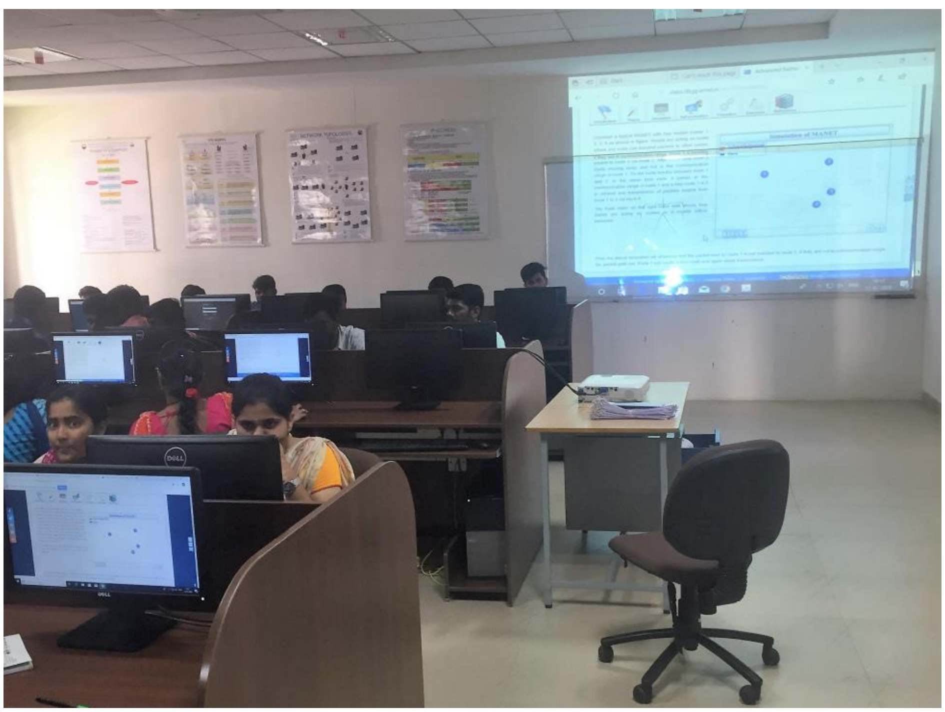
❖ Is: source address, source port number
❖ Id: dest address, dest port number
❖ It: packet type
❖ Il: packet size
❖ If: flow id
❖ Ii: unique id
❖ Iv: TTL value

Field 7:

❖ Packet info at "Application level" which consists of the type of application like ARP, TCP, and Adhoc routing protocols like DSDV, DSR, AODV etc. The field consists of a leading **–P** and the list of tags for different applications.

Then, write a awk script that take data from trace file and give the report for performance metrics such as packet delivery ratio, throughput etc.

**Glimpse**



**References:**


1. http://www.jgyan.com/ns2/awk%20scripts%20for%20ns2%20result%20analysis.php
2. https://www.isi.edu/nsnam/ns/tutorial/
3. http://nile.wpi.edu/NS/